

Korede R. Bishi

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KoredeBishi

🌐 <http://koredebishi.github.io/>

🎓 Google Scholar

Employment History

- Sep, 2022 – **Teaching Assistant: School of Computing**, Franklin College of Arts and Sciences, University of Georgia, USA.
- Courses Data Structures (CSCI 2720);
 Discrete Mathematics (CSCI 2610);
 Computer Ethics (CSCI 3030).
- Sep, 2022 – **Research Assistant:** Modeling, Simulation & Analytics Lab (MSAL), School of Computing, University of Georgia.
- Advisor Dr. John A. Miller.
- Jan, 2017 – 2022 **Distribution and Transport Operations Manager.** SevenUp Bottling Company (PepsiCo Nigeria).

Education

- 2022 – 2027 **Ph.D. in Computer Science, University of Georgia, USA.**
Advisor: Dr. John A. Miller
- 2007 – 2012 **B.Sc. in Computer Science, University of Benin, Nigeria.**

Research

Research Interests

- Areas Transportation AI; Traffic Forecasting; Digital Twins; Graph Neural Networks; Microscopic Traffic Simulation; Modeling and Simulation; Intelligent Transportation Systems

Current Research

- 2022 – **Dissertation Research: AI-Enabled Simulation for Resilient Transportation Systems.** Developing empirically validated and agentic microscopic traffic-simulation methods for transportation resilience, integrating lane-level PeMS validation and HPC calibration in ScalaTion 2.0; wildfire evacuation and contraflow evaluation; and *AutoTrafficSim*, a proposed AI-agent architecture for validated simulation experimentation.

Software & Code

- ScalaTion Microscopic traffic-simulation codebase (ScalaTion 2.0): <https://github.com/koredebishi/scalation>

Research (continued)

Time-Series AI  Foundation-model forecasting, distribution-shift stress testing, and spacecraft-telemetry anomaly detection: https://github.com/koredebishi/IEEE_BigData_2026

Honors & Awards

2026  **Overall Best Paper Runner-Up Award**, 2026 Annual Modeling and Simulation Conference (ANNSIM 2026), held in cooperation with the Association for Computing Machinery Special Interest Group on Simulation and Modeling (ACM SIGSIM).

Publications

Peer-Reviewed Publications

- 2026  K. R. **Bishi**, C. Bowman, and J. A. Miller, “Beyond Corridor Averages: Lane-Level Validation of Microscopic Freeway Simulation with Empirically Parameterized Arrivals,” in *Proceedings of the 2026 Annual Modeling and Simulation Conference (ANNSIM 2026)*, held in cooperation with the Association for Computing Machinery Special Interest Group on Simulation and Modeling (**ACM SIGSIM**). (**Overall Best Paper Runner-Up Award**.)
-  A. O. Okunoye, M. D. Hodgson, K. R. **Bishi**, and I. B. Arpinar, “RC-GNN: Robust Causal Graph Neural Networks under Compound Sensor Corruptions,” *Machine Learning from Challenging Data*, vol. 14030, p. 1403003, 2026.
-  A. O. Okunoye, D. Okonkwo, K. R. **Bishi**, and I. B. Arpinar, “GlycoLLM: Selective Classification of WURCS-Encoded Glycan Structures under Distribution Shift,” *SPIE Optics + Photonics 2026, Applications of Machine Learning 2026*, San Diego, CA, Aug. 27, 2026, Paper 14218-20.

Manuscripts Under Review

-  D. Okonkwo, A. Okunoye, K. R. **Bishi**, and I. B. Arpinar, “The Model Knows Before It Shows: Measuring, Locating, and Detecting Epistemic Spinning in Large Language Model Reasoning.” Under review at the 38th IEEE International Conference on Tools with Artificial Intelligence (ICTAI 2026).
-  K. R. **Bishi**, A. Okunoye, and D. Okonkwo, “Does Drift Severity Predict Forecaster Risk in Spacecraft Telemetry?” Under review at the 1st IEEE International Conference on Security and Resiliency of Critical Infrastructure and Space Technologies (IEEE RISC 2026).
-  K. R. **Bishi**, D. Okonkwo, and A. Okunoye, “Severe but Harmless: Stress-Testing Shift Severity for Frozen Time-Series Forecasters.” Under review at the 2026 IEEE International Conference on Big Data (IEEE BigData 2026).
- 2027  K. R. **Bishi**, A. Okunoye, and D. Okonkwo, “Training, Not Information: What Zero-Shot Time-Series Foundation Models Lack and What Fine-Tuning Buys Back.” Under review at the AAAI Conference on Artificial Intelligence (AAAI 2027).

Manuscripts in Preparation

-  K. R. **Bishi** and J. A. Miller, “Wildfire Evacuation Resilience and Contraflow Evaluation Using Microscopic Traffic Simulation.” (in preparation)
-  K. R. **Bishi** and J. A. Miller, “AutoTrafficSim: An Agentic Simulation Architecture for Validated Traffic-Simulation Experimentation.” (in preparation)

Presentations

- 2026  “Beyond Corridor Averages: Lane-Level Validation of Microscopic Freeway Simulation with Empirically Parameterized Arrivals,” 2026 Annual Modeling and Simulation Conference (ANNSIM 2026), held in cooperation with the Association for Computing Machinery Special Interest Group on Simulation and Modeling (ACM SIGSIM).

Professional Affiliations & Certifications

Professional Memberships

- Member  Institute for Operations Research and the Management Sciences (INFORMS);
 Institute of Electrical and Electronics Engineers (IEEE);
 Association for Supply Chain Management (ASCM);

Certification

- 2025  *Neural Networks and Deep Learning* — DeepLearning.AI;

 *Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization* — DeepLearning.AI;
- 2022  *AWS Certified Cloud Practitioner* — Amazon Web Services (AWS), issued Dec 2022

 *Verified International Academic Qualifications* — World Education Services (WES), issued Mar 2022

 *Certified in Logistics, Transportation and Distribution* — Association for Supply Chain Management (ASCM), Credential ID: APICS2029139.

Skills

- Machine Learning  PyTorch, scikit-learn, graph neural networks, time-series modeling
- Cloud & DevOps  AWS (EC2, IAM, VPC), Docker, CI/CD
- Data Tools  Pandas, NumPy, Matplotlib, seaborn
- Simulation  ScalaTion 2.0, SUMO, microscopic traffic simulation
- Coding  Java, Scala, Python
- Misc.  Academic research, teaching, training, and $\text{\LaTeX}$ typesetting and publishing